

Bingzhuo TIAN

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Research Interests: Operating Systems and Robotics System Software — ROS 2/DDS, control-chain latency, device communication, and system observability.

EDUCATION

- Hebei University of Technology, B.S. Candidate in Applied Physics, expected Jul 2027; GPA 3.18/4.00.
- Micro-majors: Complex Equipment Digital Twin (GPA 3.64/4.00); AI + IoT, Bilingual (GPA 4.00/4.00); coursework in machine vision, PLC, embedded systems, and intelligent sensing.
- University of Arizona, College of Science — Applied physics-related joint training / non-degree coursework, GPA 3.40/4.00; A grades in ENGL 106/107/108.

SELECTED PROJECTS

Raspberry Pi 5 Intelligent Rescue Vehicle | Main program & system integration 2024.07 – 2024.12

- Owned the main control framework and hardware-software integration, while consuming vision outputs from the recognition module; integrated IMU, motor/servo/fan/gate control, and task-state transitions.
- Worked with Python, OpenCV interfaces, Picamera2, RPi.GPIO, gpiozero, and smbus2; awarded Second Prize, Hebei Provincial Engineering Practice and Innovation Competition.

Improved Young's Modulus Measurement & Remote Preview System | Co-lead, host-side software Mar – Sep 2025; awarded Dec 2025

- Co-led the digital workflow and host-side software; implemented the C# / WinUI 3 interface and contributed to physical-structure principle design.
- Converted a manual-reading physics experiment into a data-acquisition / waveform-visualisation / report-generation / remote-preview workflow; awarded Third Prize, 11th National Undergraduate Physics Experiment Competition.

National Smart Car Competition — Vision Group Hardware | PCB design & soldering 2023.12 – 2024.07

- Designed and soldered PCB boards for the vision-group platform; supported hardware debugging and system integration; awarded Second Prize, regional contest (Vision Group A).

Postcard Digital Arrival Platform | Project Lead & Product Architect (BoHack 2026, WIIC) May 2026 (42h)

- Led product vision, system architecture (5-repo: cloud backend, PWA, Android, Python CV/AI), and OCR/AI pipeline prototyping (OpenCV + PaddleOCR + local LLM) in a 42-hour hackathon at the World Intelligent Industry Conference; delivered market research, pitch strategy, and live demo.

OPEN SOURCE & TECHNICAL COMMUNITY

- Founder & President, AI Innovation Club, Hebei University of Technology — operated a 200+ member technical community on WeLink; organised learning activities around OpenCamp, rCore/ArceOS, Linux, open-source systems, hardware, and intelligent systems.
- Outstanding University Volunteer, OpenCamp Open Source Training Program — connected operating-systems and open-source resources with campus learning. Public page: opencamp.cn/awards/2025

TECHNICAL SKILLS

- Programming: Python, C/C++, Rust, C#/.NET, R. Systems & Embedded: Linux, Git, Raspberry Pi, GPIO, I2C, serial communication, IMU, actuator control. Vision & Data: OpenCV, NumPy, Picamera2, experimental data processing & visualisation.